

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1025

Marion S 138 kV
November 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1025. The Interconnection Customer proposes a 130.0 MW battery energy storage facility interconnecting at the Marion S 138 kV substation in the Pennsylvania area. The request was received on July 09, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 368 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$77,948,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Colver Power Project to Deerhaven 138 KV	Armstrong to Ft Gibson Substation 500 KV	146.3	205	6.5
Deerhaven Substation to Oneill 138 KV	Ft Gibson Substation to Armsrong 500 KV	124.0	237	5.8
High Sheldon to Bass Lake 138 KV	Coronet to Fort Humbug Substation 500 KV	122.6	195	5.2
Edes Tap 2 to HamiltonRd 138 KV	Lutesville to Edes Tap 2 345 KV	117.3	201	6.1
Astoria Energy li to Colver Power Project 138 KV	Beardsley to Purdue Northwest 138 KV Line	115.8	280	18.8
Hamilton Rd Substation to Wien 138 KV	Lutesville to Edes Tap 2 345 KV	115.6	214	6.1
Hernando to Bass Lake 138 KV	Ft Humbug to Coronet 500 KV Line	115.1	204	5.2
Wien to Warrior Ridge Hydro 138 KV L	Edes Tap 2 to Lutesville 345 KV	112.1	223	6.1
Rollins Wind Project to Astoria Energy li 138 KV line	Beardsley to Purdue Northwest 138 KV	110.9	290	18.8
High Sheldon to Midwind 138 KV L	Coronet to Ft Hubug 500 KV line	108.5	228	5.2
Midwind to Silverton 138 KV	Coronet to FortHumbug 500 KV line	107.3	242	5.2
Hernando to Walnut Crk 138 KV	Coronet to Ft Humbug 500 KV	107.3	212	5.2
Silverton to Baja 138 KV	Edes Tap 2 to Lutesville 345 KV L	107.2	201	13.5
Imperial to Eufaula 500 KV	Trego to Galion Generating Sta 500 KV	106.3	4,273	8.9
Trego to Troy Energy Llc 500 KV	Swan Valley to Trego 500 KV line	106.1	4,121	8.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sunrise Center to Nodaway 138 KV	Edes Tap 2 to Lutesville 345 KV	105.9	231	9.9
Trego Substation to Galion Generating Sta 500 KV	Trego to Beaur 500 KV	103.7	3,288	5.3
Beaver to Eufaula 500 KV	Trego to Galion Generating Sta Substation 500 KV	103.7	5,086	9.2
Marion S to Puna Geothermal Venture I 138 KV	Edes Tap 2 Substation to Lutesville 345 KV	101.0	229	13.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Deerhaven to Colver Power Project 138 KV Line	Rebuild line with higher-capacity conductor	\$6,166,000	100% (\$6,166,000)
Deerhaven Substation to Oneill 138 KV	Rebuild line with higher-capacity conductor	\$5,307,000	100% (\$5,307,000)
High Sheldon to Bass Lake 138 KV	Rebuild line with higher-capacity conductor	\$6,404,000	60% (\$3,876,000)
Edes Tap 2 to HamiltonRd 138 KV	Rebuild line with higher-capacity conductor	\$2,056,000	40% (\$833,000)
Astoria Energy li to Colver Power Project 138 KV	Rebuild line with higher-capacity conductor	\$14,650,000	100% (\$14,650,000)
Hamilton Rd to Wien 138 KV	Rebuild line with higher-capacity conductor	\$6,570,000	40% (\$2,663,000)
Hernando to Bass Lake 138 KV	Rebuild line with higher-capacity conductor	\$9,801,000	60% (\$5,931,000)
Wien to Warrior Ridge Hydro 138 KV L	Rebuild line with higher-capacity conductor	\$7,826,000	40% (\$3,172,000)
Astoria Energy li to Rollins Wind Project 138 KV	Rebuild line with higher-capacity conductor	\$7,036,000	100% (\$7,036,000)
High Sheldon to Midwind 138 KV Line	Reconductor line	\$891,000	60% (\$539,000)
Midwind to Silverton 138 KV	Reconductor line	\$3,873,000	60% (\$2,344,000)
Hernando to Walnut Crk 138 KV	Reconductor line	\$2,142,000	60% (\$1,297,000)
Silverton to Baja 138 KV	Reconductor line	\$10,155,000	100% (\$10,155,000)
Imperial to Eufaula 500 KV	Reconductor line	\$34,305,000	1% (\$356,000)
Trego to Troy Energy Llc Substation 500 KV	Reconductor line	\$53,872,000	1% (\$555,000)
Sunrise Center to Nodaway 138 KV	Reconductor line	\$4,849,000	61% (\$2,963,000)
Trego Substation to Galion Generating Sta 500 KV	Reconductor line	\$50,189,000	1% (\$515,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beaver to Eufaula Substation 500 KV	Reconductor line	\$21,393,000	1% (\$222,000)
Marion S to Puna Geothermal Venture I 138 KV line	Reconductor line	\$9,368,000	100% (\$9,368,000)
	Total allocated to project		\$77,948,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.